

# ARJUN PANDEY

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## WORK EXPERIENCE

### Software Engineer II

June 2025 – Present

*Morgan Stanley (via Netconnect Global)*

*Bangalore, India*

- Engineered and scaled cloud-native backend services for derivatives clearing and post-trade processing platforms using **Java 21**, **Spring Boot**, and distributed messaging, supporting high-volume, low-latency financial workflows.
- Designed and optimized asynchronous, event-driven processing pipelines leveraging **Kafka**, **JMS**, and database-backed coordination to reliably process large-scale trade lifecycle events under sustained peak load conditions.
- Improved overall system throughput by approximately **50%** by implementing **JDBC batch processing**, increasing broker and consumer parallelism, and enabling **horizontal scaling** across processing nodes without compromising data consistency.
- Implemented robust persistence and data-access layers using **Spring Data JPA** and optimized SQL patterns, separating read/write workloads and reducing database contention in latency-sensitive trade calculations.
- Reduced database load on critical execution paths by introducing **in-memory and Redis-based caching strategies**, applying cache-aside patterns for frequently accessed reference and position data used in real-time computations.
- Built and exposed secure RESTful APIs using **Spring MVC**, **Spring Security OAuth2**, and OpenAPI specifications, ensuring reliable integration with internal and downstream enterprise systems.
- Instrumented end-to-end observability using centralized logging, metrics, and distributed tracing to diagnose performance bottlenecks and ensure production stability in high-throughput trading environments.
- Managed schema evolution using **Liquibase**, enabling controlled, versioned database migrations aligned with infrastructure-as-code and enterprise release governance standards.
- Collaborated with platform, QA, and operations teams to design resilient services with fault-tolerant messaging, retry handling, and graceful degradation suitable for mission-critical financial systems.

### Software Engineer

Oct 2022 – May 2025

*Yield Engineering Systems*

*Bangalore, India*

- Built a full-scale **digital twin** simulating real semiconductor hardware behavior including **wafer handling, load ports, and robotic arms**, improving system validation and testing efficiency by **50%**.
- Developed robotic integration and control services for **Kawasaki robots** and **UR10**, enabling automated wafer transfer workflows and safe testing without physical tool dependencies.
- Designed a **wafer lifecycle history database** using **JPA** and relational schemas, reducing data retrieval time by **60%** and improving traceability across tool operations.
- Automated Azure DevOps repository backups using **Java Spring Boot**, eliminating manual effort and saving approximately **4 hours per week** for the engineering team.
- Applied **SOLID principles** and design patterns to improve backend scalability and maintainability, resulting in a **30% reduction in overall development time**.
- Worked across **Java and .NET (C#)** codebases and collaborated with global clients including **Intel, Micron, Wolfspeed, TSMC, and NASA JPL** to deliver production-grade semiconductor solutions.
- Tech Stack:** Java, Spring Boot, C#, SECS/GEM, TCP/IP, MySQL, Kafka, Git, Azure DevOps, AWS (EC2, S3, Lambda)

## TECHNICAL SKILLS

**Languages:** Java, SQL, JavaScript, C++ , C#

**Familiar Libraries/Frameworks:** **Java Spring Boot**, JPA, **Microservices**, J2EE, RESTful APIs, Multithreading, Socket Programming, Message Queues (Kafka), Git, TCP/IP

**Developer Tools:** Eclipse, Postman, SQL Server Management Studio, VS Code, JIRA, Confluence

**Methodologies:** Agile (Scrum, Kanban), Design Patterns, SOLID Principles

**Coursework:** Database Management Systems (DBMS), Data Structures & Algorithms (DSA), Operating Systems, Object-Oriented Programming (OOPS)

## Achievements

- Solved **500+** DSA problems across LeetCode and HackerRank, demonstrating strong problem-solving and algorithmic skills.
- Secured **Rank 7250** in **TCS CodeVita Season 9**, placing in the **top 7%** of participants.

## EDUCATION

**Lovely Professional University**

06 2018 – 06 2022

*B.Tech in Computer Science and Engineering - 7.7 / 10*

*Punjab, India*